

Advanced EC2 Worker Node Monitoring Dashboard Guide

STEP 1 — AWS Infrastructure Setup

Create:

- 1 Control Node
- 2 Worker Nodes

Important:

- Use same PEM key for all EC2 instances
- Security Group should allow:
 - SSH Port 22
 - HTTP Port 80

Recommended OS:

Amazon Linux 2

STEP 2 — Install Required Packages

```
sudo yum update -y
sudo yum install ansible nginx python3-pip -y
pip3 install boto3 botocore
ansible-galaxy collection install amazon.aws
```

STEP 3 — Verify Installation

```
ansible --version
python3 --version
nginx -v
```

STEP 4 — Configure AWS Credentials

```
aws configure
```

Enter:

AWS Access Key ID

AWS Secret Access Key

Region = ap-south-1

Output = json

STEP 5 — Configure PEM Key

```
sudo cp dynamic.pem /etc/ansible/
```

Fix permissions:

```
sudo chown ec2-user:ec2-user /etc/ansible/dynamic.pem
```

```
sudo chmod 400 /etc/ansible/dynamic.pem
```

STEP 6 — Create Project Directory

```
mkdir -p ~/ansible/templates
cd ~/ansible
```

STEP 7 — Create Dynamic Inventory File

```
vi aws_ec2.yml

Paste:
plugin: amazon.aws.aws_ec2
regions:
  - ap-south-1
hostnames:
  - private-ip-address
filters:
  instance-state-name:
    - running
    - stopped
compose:
  ansible_host: private_ip_address
ansible_user: ec2-user
ansible_ssh_private_key_file: /etc/ansible/dynamic.pem
keyed_groups:
  - key: tags.Name
    prefix: ""
```

STEP 8 — Create ansible.cfg

```
vi ansible.cfg

Paste:
[defaults]
inventory = aws_ec2.yml
private_key_file = /etc/ansible/dynamic.pem
host_key_checking = False

[inventory]
enable_plugins = amazon.aws.aws_ec2
```

STEP 9 — Test Dynamic Inventory

```
ansible-inventory --graph
```

Expected Example:

```
@all:
  |_Work_Node_1
  |_Work_Node_2
  |_Control_Node
```

STEP 10 — Test SSH Connectivity

```
ansible _Work_Node_* -m ping
```

Expected:

```
SUCCESS => {
  "ping": "pong"
}
```

STEP 11 — Start NGINX

```
sudo systemctl enable nginx
sudo systemctl start nginx
sudo systemctl status nginx
```

STEP 12 — Create monitor.yml

```
vi monitor.yml
Paste:
```

```

---
- name: Worker Node Monitoring
  hosts: localhost
  gather_facts: yes

  tasks:
    - name: Get EC2 instance info
      amazon.aws.ec2_instance_info:
        region: ap-south-1
        register: ec2_info

    - name: Generate monitoring webpage
      become: yes
      template:
        src: templates/index.j2
        dest: /usr/share/nginx/html/index.html

```

STEP 13 — Create Dashboard Template

```

mkdir templates
vi templates/index.j2

```

Paste the advanced HTML dashboard template.

STEP 14 — Run Playbook

```

ansible-playbook monitor.yml

```

STEP 15 — Open Dashboard

```

Open browser:
http://YOUR_PUBLIC_IP

```

STEP 16 — ENABLE AUTO REFRESH

Open crontab:

```

sudo yum install crontie -y

```

```

crontab -e

```

Add this line:

```

*/1 * * * * export AWS_SHARED_CREDENTIALS_FILE=/home/ec2-user/.aws/credentials && cd /etc/ansible && /usr/bin/a

```

Save and exit.

Then ensure cron service is running:

```

sudo systemctl enable crond
sudo systemctl start crond
sudo systemctl status crond

```

Now verify after 1 minute:

```

cat /home/ec2-user/cron.log

```

STEP 17 — Dashboard Features

Your dashboard now supports:

- Running / Stopped status
- Search bar

- Status filter
- Summary cards
- Public & Private IP
- Availability Zone
- Auto refresh every 60 seconds
- Modern dark UI

STEP 18 — Common Errors and Fixes

1. boto3 not found
Fix:
`sudo pip3 install boto3 botocore`
2. Permission denied for nginx html directory
Fix:
Use become: yes only for template task
3. ansible_date_time undefined
Fix:
Use gather_facts: yes
4. launch_time timestamp error
Fix:
Use:
`{{ reservation.launch_time[:16] }}`
5. Unable to locate credentials
Fix:
Run playbook using ec2-user, not sudo
6. Dashboard not opening
Fix:
Allow HTTP port 80 in Security Group

STEP 19 — Future Improvements

- You can later add:
- CPU monitoring
 - RAM monitoring
 - Disk usage
 - Docker container health
 - Jenkins pipeline status
 - Prometheus metrics
 - Grafana integration
 - Slack/Telegram alerts

STEP 20 — Resume Description

Built an automated EC2 worker node monitoring dashboard using Ansible, AWS Dynamic Inventory, Jinja2 templates, and NGINX. Implemented real-time infrastructure visibility with active/stopped instance monitoring, search/filter functionality, and automated webpage generation.

Congratulations! You have successfully created a real-world DevOps monitoring project using AWS, Ansible, NGINX.